

IoT System Design and Analysis

Project Instructions:

You are required to design a simple Internet of Things (IoT) system. Follow the steps below to complete the project. Use clear explanations, diagrams, and calculations as needed.

Step 1: Choose Your IoT Topic

- Pick an IoT application area

Step 2: Identify Objects

- List the **General Object** (main system or environment).
- Identify **Specific Objects** (devices or components used in your system).

Step 3: Lecture Review and Identify Components

- Review your lecture notes on IoT components.
- Identify and list the key components your system will use:
 - Sensors
 - Microcontroller or processing unit
 - Connectivity module
 - Actuators (if any)
 - Power supply

Step 4: Calculate Power Consumption

- Estimate the power consumption of each component (use datasheets or typical values).
- Calculate total power consumption of the system.
- Example:
 - Sensor: 0.1W
 - Microcontroller: 0.5W
 - WiFi module: 1W
 - Total: 1.6W

Step 5: Create a Flowchart

- Draw a simple flowchart showing how your IoT system works.
- Include steps like sensing, data transmission, processing, and action.

Step 6: Write a Conclusion

- Summarize what your IoT system does.
- Explain benefits and possible challenges.
- Suggest improvements or future extensions.

Submission Guidelines:

- Use clear, simple language.
- Include diagrams and calculations.
- Keep the project concise and well-organized.

- Submit your completed project via email
to: **joseptuyish@gmail.com**

Good luck! Be creative and apply what you have learned.